Chapter 25

Stakeholders, Resources, and Documents in Clinical Research

Deepak C. Chilkoti
Consultant, Clinical Operations and Pharmacovigilance, Delhi, India

Learning Objectives:

Objective 25.1 Explain responsibilities of stakeholders in clinical research.
Objective 25.2 Overview of the common resources and documents used in clinical research.

OBJECTIVE 25.1. EXPLAIN RESPONSIBILITIES OF STAKEHOLDERS IN CLINICAL RESEARCH

Stakeholders in clinical research have the responsibility to ensure that all the "thirteen" principles of the International Conference on Harmonization Good Clinical Practice (ICH-GCP) are complied with. The Institutional Review Board/Independent Ethics Committee (IRB/IEC), investigator, and sponsor are the key stakeholders, and other stakeholders such as the regulator, contract research organization (CRO), hospital/clinic, auditor, clinical research coordinator (CRC)/clinical research associate, data and safety monitoring board, subjects, etc. also have important roles during the lifecycle of trial conduct. Most of the following contents under objective 25.1 are extracted from ICH-GCP.[1]

Institutional Review Board/Independent Ethics Committee

IRB/IEC members evaluate the medical, nonmedical, ethical, and legal aspects of trials by collective qualifications and experience of a reasonable number of members (at least five).

Responsibilities

- Rights, safety, and well-being of trial subjects especially those trials that include vulnerable subjects should be safeguarded.
- The IRB/IEC should review the following documents to fulfill its responsibilities: trial protocol(s)/amendment(s), written informed consent form(s), investigator's brochure (IB), subject recruitment procedures (e.g., advertisements), written information to be provided to subjects, information about payments and compensation available to subjects, the investigator's current curriculum vitae, etc.
- Review of a proposed clinical trial by the IRB/IEC should happen within a reasonable time, and the IRB should document its views in writing, clearly identifying the trial and the documents reviewed. IRB minutes should mention the following:
  - approval;
  - modifications required before its approval;
  - disapproval; and
  - termination/suspension of any prior approval.
- The IRB/IEC should review the qualifications and prior experience of the investigator for the proposed trial, as documented by a current curriculum vita before according approval to the investigator.

Clinical Pharmacy Education, Practice and Research. https://doi.org/10.1016/B978-0-12-814276-9.00025-8
Copyright © 2019 Elsevier Inc. All rights reserved.

- The IRB/IEC to ensure that the safety of human subjects is not jeopardized; it should conduct a continuing review of each ongoing trial at intervals depending on the degree of risk, but at least once per year.
- In emergency situations wherein prior consent from the trial subject or the subject's legally acceptable representative cannot be obtained, the IRB/IEC should ensure that the trial documents meet regulatory requirements for such trials and address ethical concerns.
- The IRB/IEC should ensure that, on the informed consent form, information regarding payment to subjects is mentioned. The IRB/IEC should review the amount and method of payment to subjects to assure that neither presents problems of coercion or undue influence on the trial subjects. Furthermore, payments should be made on a prorated basis and not on completion of the trial by the subject.

Investigator

The investigator should comply with the Good Clinical Practice (GCP) and the applicable regulatory requirements during study conduct. The investigator conducting the trial should be qualified by education, training, and experience and should provide evidence of such qualifications through curriculum vitae and/or other relevant documentation. The investigator should be thoroughly familiar with the use of the investigational product, as described in the study documents (protocol, IB, and product information leaflet) provided by the sponsor. The investigator should permit monitoring and auditing by the sponsor, IRB/IEC, and regulatory authority. The investigator should maintain a list of qualified persons to whom the investigator has delegated significant trial-related duties.

Responsibilities of the Investigator

Adequate Resources for the Conduct of the Trial

- The investigator should be able to demonstrate the potential for recruiting suitable subjects within project timelines.
- The investigator should have sufficient time to properly conduct and complete the trial within the agreed trial period.
- The investigator should have an adequate number of qualified staff and adequate facilities for the duration of the trial.
- The investigator should ensure that the site team assisting with the trial is adequately informed about the protocol, the investigational product(s), and their trial-related duties and functions.

Communication With the IRB/IEC

- Before initiating a trial, the investigator should have written and dated approval from the IRB/IEC for the trial protocol, written informed consent form, and subject recruitment procedures (e.g., advertisements).
- The investigator should provide the IRB/IEC with a current/updated copy of the IB.
- During the trial, the investigator should submit to the IRB/IEC all documents subject to review for approval (e.g., consent updates).

Medical Care of Trial Subjects

- For all trial-related medical decisions, a qualified physician, who is an investigator or a subinvestigator for the trial, should be responsible.
- The investigator should inform the subject's primary physician about the subject's participation in the trial if the subject agrees to the primary physician being informed.
- If a subject decides to withdraw prematurely from a trial, the subject is not obliged to give a reason, and the investigator should make all efforts to know the reason while fully respecting subject's rights.

Compliance With Protocol

- The investigator should conduct the trial in compliance with the protocol agreed to by the sponsor, by the regulatory authority, and which was given approval by the IRB/IEC.
- The investigator should not implement any changes of the protocol without agreement by the sponsor and prior review and documented approval from the IRB/IEC of an amendment, except where necessary to eliminate an immediate hazard(s) to trial subjects, or when the change(s) involves only logistical or administrative aspects of the trial (e.g., change in monitor(s), change of telephone number(s)).

Investigational Product(s)

- Responsibility rests with the investigator for investigational product(s) accountability at the trial site. The investigator may assign responsibility for investigational product(s) accountability to a pharmacist who is under the supervision of the investigator.
- The investigational product should be stored as specified by the sponsor and by the applicable regulatory requirement.
- The investigator should ensure that the investigational product is used only in accordance with the approved protocol.
- The investigator, or a person designated by the investigator, should explain the correct use of the investigational product to each subject.

Randomization Procedures and Unblinding

The investigator should follow the trial's randomization procedures, if any, and should ensure that the code is broken only in accordance with the protocol. In case of premature unblinding in a blinded trial due to a serious adverse event or accidental unblinding, the investigator should promptly document and explain it to the sponsor.

Informed Consent of Trial Subjects

Informed consent from trial subjects should be obtained and documented on the IRB/IEC's approved informed consent form. The investigator should adhere to ethical principles having their origin in the Declaration of Helsinki during obtainment of informed consent.

The written informed consent form should be revised whenever important new information becomes available that may be relevant to the subject's consent. Any revised written informed consent form should receive the IRB/IEC's approval opinion in advance of use.

Records/Reports and Progress Reports

- The investigator should maintain adequate and accurate source documents and trial records of the trial subjects. Source data should be attributable, legible, contemporaneous, original, accurate, and complete. Changes to source data should be dated, initialed, traceable, should not obscure the original entry, and should be explained; this applies to both written and electronic changes.
- The investigator should maintain all the essential documents for the conduct of the clinical trial and should take measures to prevent the accidental or premature destruction of these documents.
- Essential documents should be retained until at least 2 years after the last approval of a marketing application in an ICH region and until there are no pending or contemplated marketing applications in an ICH region or until at least 2 years have elapsed since the formal discontinuation of clinical development of the investigational product. These documents should be retained for a longer period, however, if required by the applicable regulatory requirements or by an agreement with the sponsor. It is the responsibility of the sponsor to inform the investigator/institution as to when these documents no longer need to be retained.
- The investigator should make available for direct access all requested trial-related records upon request of the monitor, auditor, IRB/IEC, or regulatory authority.
- The investigator should submit written progress reports of the trial status to the IRB/IEC annually.

Safety Reporting

- All serious adverse events (SAEs) should be reported immediately to the sponsor except those SAEs that the protocol identifies as not needing immediate reporting. The immediate reports should be followed promptly by detailed, written reports.
- Adverse events and/or laboratory abnormalities identified in the protocol as critical to safety evaluations should be reported to the sponsor.

Premature Termination or Suspension of a Trial

- If the trial is prematurely terminated or suspended for any reason, the investigator should promptly inform the trial subjects and should assure follow-up and appropriate therapy for the subjects.

Final Report(s) by the Investigator

- Upon completion of the trial, the investigator should inform the institution and should provide the IRB/IEC with a summary of the trial's outcome, and to the regulatory authority, where applicable, with any reports required with the help of sponsor.

Sponsor

The sponsor is an individual, company, institution, or organization that takes responsibility for the initiation, management, and/or financing of a clinical trial.

Responsibilities of the Sponsor

Quality Assurance and Quality Control

- The sponsor is responsible for implementing and maintaining quality assurance and quality control systems with a written standard operating procedure (SOP) to ensure that trials are conducted, and data are generated, documented, and reported in compliance with the protocol, SOPs, GCP, and the applicable regulatory requirement. Quality control should be applied to each stage of data handling to ensure that all generated data are reliable and have been processed correctly.

Contract Research Organization Selection

- A sponsor may transfer any or all of its trial-related duties and functions that should be specified in writing to a CRO but should ensure continuous oversight of activities done by the CRO, including those that are subcontracted to another party by the CRO. The ultimate responsibility for the quality and integrity of the trial data always resides with the sponsor.

Medical Expertise

- The sponsor should designate appropriately qualified medical personnel internally or appoint outside consultant to advise on trial-related medical questions.

Trial Design

- The sponsor should utilize qualified individuals (e.g., biostatisticians, clinical pharmacologists, and physicians) as appropriate, throughout all stages of the trial process, from designing the protocol and case report forms (CRFs) and planning the analyses to analyzing and preparing interim and final clinical trial reports.

Trial Management, Data Handling, and Record Keeping

- The sponsor should utilize appropriately qualified individuals to supervise the overall conduct of the trial, to handle the data, to verify the data, to conduct the statistical analyses, and to prepare the trial reports.

Investigator Selection

- The sponsor is responsible for selecting the investigator/institution. The sponsor and the investigator/institution should sign the protocol, or an alternative document (e.g., clinical trial agreement that can be bipartite between sponsor and investigator or tripartite between the sponsor, investigator, and institution), to confirm this agreement.

Allocation of Responsibilities

- Before initiating a trial, the sponsor should define, establish, and allocate all trial-related duties and functions within the organization or with external parties (CRO), if involved.

Compensation to Subjects and Investigators

- The sponsor should provide insurance or should indemnify (legal and financial coverage) the investigator/institution against claims arising from the trial, except for claims that arise from malpractice and/or negligence.
- The sponsor's policies and procedures should address the costs of treatment of trial subjects in the event of trial-related injuries.

Financing

- The financial aspects of the trial should be documented in an agreement (e.g., clinical trial agreement) between the sponsor and the investigator/institution.

Notification/Submission to Regulatory Authority(ies)

- Before initiating the clinical trial, the sponsor should submit any required application to the appropriate authority for review, acceptance, and/or permission (as required by the applicable regulatory requirement) to begin the trial.

Confirmation of Review by the IRB/IEC

The sponsor should obtain the following from the investigator/institution:

- The name and address of the investigator's/institution's IRB/IEC.
- A statement obtained from the IRB/IEC that it is organized and operates according to GCP and the applicable laws and regulations.
- Documented IRB/IEC approval of a current copy of the protocol, written informed consent form(s) and any other written information to be provided to subjects, subject recruiting procedures, and documents related to payments and compensation available to the subjects.

Information on Investigational Product(s)

- Before testing the product in the clinical phase, the sponsor should ensure that sufficient safety and efficacy data from nonclinical studies and/or clinical trials are available to support human exposure by the route, at potential dosages, for the duration, and in the trial population to be studied.
- The sponsor should update the IB as significant new information becomes available but not later than 1 year from the initial finalization date of the IB.

Manufacturing, Packaging, Labeling, and Coding Investigational Product(s)

- The sponsor should ensure that the investigational product(s) (including an active comparator(s) and placebo, if applicable) is manufactured in accordance with any applicable good manufacturing practices and is coded and labeled in a manner that protects the blinding, if applicable. In addition, the labeling should comply with the applicable regulatory requirement.
- The sponsor should determine, for the investigational product(s), acceptable storage temperatures, storage conditions (e.g., protection from light), storage times, reconstitution fluids and procedures, and devices for product infusion, if any. The sponsor should inform all involved parties (e.g., monitors, investigators, pharmacists, storage managers) of these determinations.
- The investigational product(s) should be packaged to prevent contamination and unacceptable deterioration during transport and storage.
- In blinded trials, the coding system for the investigational product(s) should include a mechanism that permits rapid identification of the product(s) in case of a medical emergency but does not permit undetectable breaks of the blinding.
- If significant formulation changes are made in the investigational or comparator product(s) during the course of clinical development, the results of any additional studies of the formulated product(s) (e.g., stability, dissolution rate, bioavailability) should be available before the use of the new formulation in clinical trials. Additional studies are needed to assess whether these changes would significantly alter the pharmacokinetic profile of the product.

Supplying and Handling Investigational Product(s)

- The sponsor is responsible for supplying the investigator with the investigational product only after obtaining approval from the IRB/IEC and regulatory authority.
- The sponsor should ensure that written procedures for investigational products address adequate and safe receipt, handling, storage, dispensing, retrieval of unused product from subjects, and the return of the unused investigational product to the sponsor (or alternative disposition at the investigator site if authorized by the sponsor).
- The sponsor should
  - ensure timely delivery of investigational product to the investigator;
  - maintain records that document shipment, receipt, disposition, return, and destruction of the investigational product;

- maintain a system for retrieving investigational products and documenting this retrieval (e.g., for deficient product recall, reclaim after trial completion, expired product reclaim);
- maintain a system for the disposition of unused investigational product and for the documentation of this disposition;
- take steps to ensure that the investigational product(s) are stable over the period of use;
- maintain sufficient quantities of the investigational product(s) used in the trials to reconfirm specifications, should this become necessary, and maintain records of batch sample analyses and characteristics. To the extent stability permits, samples should be retained either until the analyses of the trial data are complete or as required by the applicable regulatory requirement, whichever represents the longer retention period.

Record Access

- The sponsor should ensure that it is specified in the protocol or other written agreement (e.g., clinical trial agreement) that the investigator/institution should provide direct access to source data/documents for monitoring, audits, IRB/IEC review, and regulatory inspection.
- The sponsor should verify that each subject has consented, in writing, to direct access to their original medical records for trial-related monitoring, audit, IRB/IEC review, and regulatory inspection.

Safety Information and Adverse Drug Reaction Reporting

- The sponsor is responsible for the ongoing safety evaluation of the investigational product.
- The sponsor should promptly notify all concerned investigators/institutions and the regulatory authority of findings that could affect the safety of subjects or alter the IRB/IEC's approval opinion to continue the trial.
- The sponsor should expedite reporting of all adverse drug reactions that are both serious and unexpected to all concerned investigator(s)/institutions(s), to the IRB(s)/IEC(s), where required, and to the regulatory authority.
- The sponsor should submit all safety updates and periodic reports to the regulatory authority, as required by applicable regulatory requirement.

Monitoring

- The sponsor should develop a systematic, prioritized, risk-based approach to monitoring clinical trials. The sponsor may choose onsite monitoring, a combination of onsite and centralized monitoring, or, where justified, centralized monitoring. The sponsor should document the rationale for the chosen monitoring strategy (e.g., in the monitoring plan). Onsite monitoring is performed at the sites at which the clinical trial is being conducted. Centralized monitoring is a remote evaluation of accumulating data, performed in a timely manner, supported by appropriately qualified and trained persons (e.g., data managers, biostatisticians). Centralized monitoring processes provide additional monitoring capabilities that can complement and reduce the extent and/or frequency of onsite monitoring and help distinguish between reliable data and potentially unreliable data.

Audit

- Sponsors should perform audits as part of implementing quality assurance. The purpose of a sponsor's audit, which is independent of and separate from routine monitoring or quality control functions, should be to evaluate trial conduct and compliance with the protocol, SOPs, GCP, and the applicable regulatory requirements.

Premature Termination or Suspension of a Trial

- If a trial is prematurely terminated or suspended, the sponsor should promptly inform the investigators/institutions, IRB/IEC, and the regulatory authority of the reason(s) for the termination or suspension.

Clinical Trial/Study Reports

- The sponsor should ensure that the clinical trial reports are prepared and provided to the regulatory agency whether the trial is completed or prematurely terminated. The sponsor should also ensure that in marketing applications the clinical trial reports should meet the standards of the ICH Guideline for Structure and Content of Clinical Study Reports.

Regulator

The regulatory authority (e.g., Food and Drug Administration [FDA], Medicines and Healthcare Products Regulatory Agency) is in charge of protecting public health by assuring the safety and efficacy of drugs, biologics, and medical devices. The regulator achieves this through an extensive review process. In this process, the sponsor works with the regulator to submit an application or investigational new drug/new drug application for the new product. The regulator carefully evaluates the submitted documents, which then leads to either approval or rejection. In addition to the review process, the regulator also performs audits and inspections of any section of clinical research, including sites, sponsors, or IRBs. If the regulator discovers any problems of noncompliance, a warning letter may be sent mentioning corrective action that must be taken immediately by the stakeholder responsible.[2]

Contract Research Organization

A CRO is a person or an organization (commercial, academic, or other) contracted by the sponsor to perform one or more of a sponsor's trial-related duties and functions.

A sponsor may transfer any or all of the sponsor's trial-related duties and functions to a CRO, any trial-related duty and function that is transferred to and assumed by a CRO should be specified in writing.

All responsibilities of a sponsor in this chapter also apply to a CRO to the extent that a CRO has assumed the trial-related duties and functions of a sponsor.

Auditor

The general responsibilities of an auditor include the following:

- Coordinate and conduct internal system audits and external investigative site/vendor audits;
- Create, maintain, and revise departmental standard operating procedures, forms, and templates;
- Host audits by sponsors and regulatory inspectors;
- Coordinate, conduct, and track company-wide regulatory training (ICH-GCP and medical device);
- Develop training materials and applicable tests and guides.

Clinical Research Coordinator

The general responsibilities of the CRC include the following:

- The clinical research coordinator, or CRC, is an essential member of the research team, the stakeholder who conducts the study. This member plays a critical role in organizing a site's participation in the expanding arena of clinical trials.
- The primary function of the CRC is to manage and coordinate the smooth, accurate progress of clinical protocols from the planning stage through study completion by acting as a liaison to other member stakeholders, including the investigator, the subject, the research site, the IRB, and the sponsor. The reason why this role holds such great importance is because the CRC is at the center of the study process.
- The CRC assists in the development, review, and maintenance of conducting a study in accordance with appropriate SOPs, GCP, and regulatory guidelines. This includes assisting in the review of protocols, checking for protocol feasibility, preparing documents for IRB submission, recruiting and screening study participants, obtaining informed consent from study subjects, facilitating continued participation of subjects, and tracking study progress while maintaining compliance.[3]

Clinical Research Associate

General responsibilities of a CRA include the following:

- Coordinating with the ethics committee, which safeguards the rights, safety, and well-being of all trial subjects;
- Managing regulatory authority applications and approvals that oversee the research and marketing of new and existing drugs;
- Identifying and assessing the suitability of facilities to be used as the clinical trial site;
- Identifying/selecting an investigator who will be responsible for the conduct of the trial at the trial site;
- Liaising with doctors/consultants or investigators on conducting the trial;
- Setting up the trial sites, which includes ensuring each center has the trial materials, including the trial investigational medicinal product. Moreover, training the site staff to trial-specific industry standards;

- Monitoring the trial throughout its duration, which involves visiting the trial sites on a regular basis;
- Verifying that data entered onto the CRFs are consistent with patient clinical notes, source data/document verification;
- Collecting completed CRFs from hospitals and general practices;
- Writing visit reports and filing and collating trial documentation and reports;
- Ensuring all unused trial supplies are accounted for;
- Closing down trial sites on completion of the trial;
- Archiving study documentation and correspondence.

Data and Safety Monitoring Board (Independent Data-Monitoring Committee)

A data safety monitoring board or independent data-monitoring committee (IDMC) that may be established by the sponsor to assess at intervals the progress of a clinical trial, the safety data, and the critical efficacy end points and to recommend to the sponsor whether to continue, modify, or stop a trial. The IDMC should have written operating procedures and maintain written records of all its meetings.

Participants (Subject/Trial Subject)

An individual who participates in a clinical trial, either as a recipient of the investigational product(s) or as a control. General responsibilities of participants include the following:[4]

- Respect investigators, research staff, and other participants.
- Read the consent form and other documents. Ask questions if they do not understand something about the study or their rights and responsibilities as a research participant or need more information.
- Carefully weigh the risks and benefits when deciding whether to participate in the study.
- Refrain from signing the consent document until they believe that they understand its content and feel comfortable with their decision to participate.
- Follow directions for proper use, dosing, and storage of self-administered study medications, providing biological samples and preparing for tests, procedures, or examinations.
- Follow directions for abstaining from nonstudy-related medications or other contraindicated medications or procedures.
- Know when the study begins and ends. This is particularly important for an intervention trial that has a follow-up period after the intervention is completed.
- Show up at scheduled appointments on time and inform the staff within a reasonable time if they need to reschedule an appointment.
- Provide truthful answers to questions asked during screening/enrollment and during the study.
- Inform staff if other medical care is needed during the study.
- Inform the staff if there are questions they would rather not answer.
- Report pain, discomfort, nausea, dizziness, and other problems and symptoms they experience during the study.
- Keep information about the study confidential, if asked to do so.
- Keep staff informed when contact information (e.g., phone number, address) changes.
- If they decide to withdraw from the study, inform the staff, and follow the procedures for withdrawal.

OBJECTIVE 25.2. OVERVIEW OF THE COMMON RESOURCES AND DOCUMENTS USED IN CLINICAL RESEARCH

GCP is an international ethical and scientific quality standard for the design, conduct, performance, monitoring, auditing, recording, analyses, and reporting of clinical trials. It also serves to protect the rights, integrity, and confidentiality of trial subjects. It is very important to understand the background of the formation of the ICH-GCP guidelines as this, in itself, explains the reasons and the need for doing so. Most of the information collected for this objective is from the ICH guidelines.[1] Today, the ICH-GCP guidelines are used in clinical trials throughout the globe with the main aim of protecting and preserving human rights. Adherence to the principles of GCPs, including adequate human subject protection, is universally recognized as a critical requirement to the conduct of research involving human subjects.[5] The development of new medicines to improve life is a long and sophisticated process involving many people and significant investments. The ICH-GCP has managed to cope with the challenges of clinical research to a major extent. The conducting of clinical research in accordance with the principles of GCP helps to ensure that the participants in clinical research are not exposed

to undue risk and that data generated in the research are valid and accurate. Thus, the GCP not only serves the interest of clinicians and those involved in the research process but also protects the rights, safety, and well-being of subjects and ensures that investigations are scientifically sound and advance public health goals.

Resources for GCP:

- ICH E6 R2 guidelines: www.ich.org/fileadmin/Public_Web_Site/ICH.../Guidelines/.../E6/E6_R2__Step_4.pdf;
- FDA website—Clinical Trials and Human Subject Protection and Office of Good Clinical Practice: https://www.fda.gov/scienceresearch/specialtopics/runningclinicaltrials/default.htm;
- Office of Good Clinical Practice's (OGCP's) mission statement;
- Online Web training by various companies and institutes;
- Colleges and institutes providing clinical research courses—various universities and institutes.

Documents in Clinical Trials

Documents are an essential part of the clinical trial where every activity performed in a clinical trial is captured. These documents are filed timely at the sites of investigator and sponsor as they aid in the proper management of a trial and provide evidence that the conduct of the clinical trial is compliant with the ICH-GCP and all applicable regulatory requirements. Data are recorded in these documents timely, which are usually audited by the sponsor and inspected by the regulatory authority to confirm the integrity of collected data and validity of the trial conducted. A trial master file is established at the beginning of the trial at the investigator's/institution's site and at the sponsor's office where all the documents are placed promptly and maintained throughout the conduct of the trial. Before the final close out of the trial, the monitor reviews both investigator/institution and sponsor files to confirm all documents are placed in appropriate files. These documents should be available for sponsor audits and regulatory inspections.

The sponsor and investigator/institution should record the location where all essential documents, including source documents, are placed. The storage system used during the trial and for archiving (irrespective of the type of media used) should provide for document identification, version history, search, and retrieval. These documents should be supplemented or reduced (if not important and relevant), in advance of trial initiation.

The sponsor should not have control of the CRF data exclusively and should ensure that the investigator has control and continuous access to the data reported to the sponsor. The copy used to replace an original document (e.g., source documents, CRF) should fulfill the requirements for certified copies. The investigator/institution should have control of all essential documents and records generated by the investigator/institution before, during, and after the trial. Details about documents are extracted mainly from the ICH-GCP.[1]

Investigator's Brochure

The IB is the compiled information (data) of clinical and nonclinical studies on the investigational product(s), which is related to the study of the investigational product(s) in human subjects. This brochure contains the relevant and current scientific information that is provided to the investigator and other members involved in the trial before initiation of the clinical trial. This information may aid in recognizing and treating possible overdose and adverse drug reactions.

The purpose is to thoroughly familiarize the investigator with the appropriate use of the investigational product(s), its possible risks and adverse reactions, specific tests, observations, and precautions that may be needed for a clinical trial. A copy of the updated IB is also provided to the IRB/IEC.

The IB is updated whenever any new significant information is available, and the same should be provided to the investigator and IRB/IEC.

Contents of the IB:

- Table of contents
- Summary
- Introduction
- Physical, chemical, and pharmaceutical properties and formulation
- Nonclinical studies
  - Nonclinical pharmacology
  - Pharmacokinetics and product metabolism in animals
  - Toxicology

- Effects in humans
  - Pharmacokinetics and product metabolism in humans
  - Safety and efficacy
  - Marketing experience
- Summary of data and guidance for the investigator
- Confidentiality statement (optional)
- Signature page (optional)

Study Protocol

Study protocol is a document describing the objective(s), design, methodology, statistical considerations, and organization of a trial providing the background and rationale for the trial. The protocol is shared with the investigator to make him/her thoroughly familiar with the appropriate use of the investigational product(s), which is then signed by the investigator for agreement with the sponsor.

The protocol is shared with the IRB/IEC for approval/favorable opinion and is also explained to other responsible members assisting in the trial. The whole clinical trial has to be conducted in compliance with the approved protocol. If there is any noncompliance with the protocol in the study, it is explained and documented by the designated person.

Contents of a trial protocol:

- General information
- Background information
- Trial objectives and purpose
- Trial design
- Selection and withdrawal of subjects
- Treatment of subjects
- Assessment of efficacy
- Assessment of safety
- Statistics
- Direct access to source data/documents
- Quality control and quality assurance
- Ethics
- Data handling and record keeping
- Financing and insurance
- Publication policy
- Supplements

Informed Consent Form

Informed consent is a process in which a subject is informed of each and every aspect of a clinical trial for which the subject confirms his/her voluntary participation. The informed consent form is the document where every aspect of the trial is written, which is signed and dated by the subject voluntarily and the form is documented along with any written information provided to the subject. A copy of the signed and dated written informed consent form and any other written information is provided to the subjects before and during conduct of the trial.

Before conduct of the trial, the IRB/IEC's written approval/favorable opinion is received on the informed consent form (including all translations).

If new information related to the subject's consent is available, the informed consent form should be revised and the written approval/favorable opinion of the IRB/IEC should be taken in advance. The subject or the subject's legally acceptable representative should be informed in a timely manner for the revised information.

The informed consent form must be in a language that is easy to understand by the subject and should be signed before participating in the trial. An impartial witness is required during the discussion of informed consent if a subject or legally acceptable representative is unable to read. The witness has to attest the information in the consent form, and any other written information is correctly explained to the subject or the subject's legally acceptable representative, and apparently understood and freely given by the subject or the subject's legally acceptable representative.

The following should be explained to subjects in informed consent discussion:

- The trial involves research.
- Purpose of the trial.
- The trial treatment(s) and the probability for random assignment to each treatment.

- The trial procedures to be followed, including all invasive procedures.
- The subject's responsibilities.
- Experimental aspects of the trial.
- The reasonably foreseeable risks or inconveniences to the subject and, when applicable, to an embryo, fetus, or nursing infant.
- The reasonably expected benefits. When there is no intended clinical benefit to the subject, the subject should be made aware of this.
- The alternative procedure(s) or course(s) of treatment that may be available to the subject, and their important potential benefits and risks.
- The compensation and/or treatment available to the subject in the event of trial-related injury.
- The anticipated prorated payment, if any, to the subject for participating in the trial.
- The anticipated expenses, if any, to the subject for participating in the trial.
- The subject's participation in the trial is voluntary and that the subject may refuse to participate or withdraw from the trial, at any time, without penalty or loss of benefits to which the subject is otherwise entitled.
- That the monitor(s), the auditor(s), the IRB/IEC, and the regulatory authority(ies) will be granted direct access to the subject's original medical records for verification of clinical trial procedures and/or data. Access should be without violating the confidentiality of the subject to the extent permitted by the applicable laws and regulations and that, by signing a written informed consent form, the subject or the subject's legally acceptable representative is authorizing such access.
- The records identifying the subject will be kept confidential and, to the extent permitted by the applicable laws and/or regulations, will not be made publicly available. If the results of the trial are published, the subject's identity will remain confidential.
- That the subject or the subject's legally acceptable representative will be informed in a timely manner if information becomes available. The information may be relevant to the subject's willingness to continue participation in the trial.
- The person(s) to contact for further information regarding the trial and the rights of trial subjects, and whom to contact in the event of trial-related injury.
- The foreseeable circumstances and/or reasons under which the subject's participation in the trial may be terminated.
- The expected duration of the subject's participation in the trial.
- The approximate number of subjects involved in the trial.

Case Report Form

A CRF, printed or electronic, is designed to collect patient data in a clinical trial; its development represents a significant part of the clinical trial and can affect study success. Site personnel capture the subject's data on the CRF, which is collected during their participation in a clinical trial. The International Conference on Harmonization Guidelines for Good Clinical Practice defines the CRF as a printed, optical, or electronic document designed to record all the protocol-required information to be reported to the sponsor on each trial subject.

Electronic CRFs are largely being used over paper CRFs due to the advantages they offer such as improved data quality, online discrepancy management, faster database lock, etc. Standard guidelines should be followed while designing any format of the CRF. A CRF completion manual should be provided to the site personnel to promote accurate data entry by them. These measures will result in reduced query generations and improved data integrity. A well-designed CRF should represent the essential contents of the study protocol and in an ideal situation; CRF is designed once the study protocol is finalized.[6]

The following points are to be borne in mind while designing a CRF:

- Use consistent formats, font style, and font sizes throughout the CRF booklet
- Select portrait versus landscape versus combination layouts
- Use clear and concise questions, prompts, and instructions
- Visual cues, such as boxes that indicate place and format of data to be recorded, should be provided to the person recording the data as much as possible
- Using the option of "circling of answers" should be limited as it is hard to interpret; instead checkboxes would be appropriate
- Clear guidance about skip patterns like what to skip and what not to skip should be mentioned at appropriate places
- Skips (instructions provided on the CRF page to maintain the connectivity between pages) should be kept to a minimum by the placement of questions to avoid confusions

- Provide boxes or separate lines to hold the answers. This indirectly informs the data recorder where to write/enter the response and helps to differentiate it visually from the entry fields for other questions
- Separate the columns with thick lines
- Provide bold and italicized instructions
- Minimize free-text responses
- Position only specified density of questions on each page
- Page numbering if necessary should be consistent throughout
- Avoid using "check all that applies" as it forces assumptions about the clinical data
- Specify the unit of measurement
- Indicate the number of decimal places to be recorded
- Use standard data format (e.g., dd/mm/yyyy) throughout the CRF
- Use precoded answer sets such as yes/no, male/female, a method of administration of medicine, and severity of adverse event (AE) (mild/moderate/severe) wherever possible
- Do not split modules/sections (a set of one or more related groups of questions that pertain to a single clinical study visit) like, for example, AE section should not be split and laid across pages such that information related to a single AE will have to be collected from different pages
- Use "no carbon required (NCR)" copies to ensure exact replica of the CRF (for paper CRFs)
- Use instructions, including page numbers where data have to be entered (e.g., during a follow-up visit, the investigator is supposed to record whether any AE has occurred and if so, details of the AE have to be recorded in the AE module. Hence, the field corresponding to this question on the module for the particular visit would have the options "yes" or "no." There should be an instruction "if 'yes,' please provide the information on the AEs page (page no. XX)."[6]

Clinical Study Report

A clinical study report is a written description of a trial/study of any therapeutic, prophylactic, or diagnostic agent conducted in human subjects, in which the clinical and statistical description, presentations, and analyses are fully integrated. This is the final report submitted to the regulatory authority with document results and interpretation of the trial. The sponsor should prepare and provide the clinical study reports (as per the standards of the ICH guideline) to the regulatory authority as required by the authority after completion or termination of the trial.[7]

List of contents in clinical study report:

- Title page
- Synopsis
- Table of contents for the individual clinical study report
- List of abbreviations and definitions of terms
- Ethics
- Independent Ethics Committee or Institutional Review Board
- Ethical conduct of the study
- Patient information and consent
- Investigators and study administrative structure
- Introduction
- Study objectives
- Investigational plan
- Study patients
- Efficacy evaluation
- Safety evaluation
- Discussion and overall conclusions
- Tables, figures, and graphs referred to but not included in the text
- Reference list
- Appendices[7]

Clinical Trial Agreement

The financial aspect of the trial should be documented in an agreement between the involved parties, e.g., the sponsor and the investigator/institution, CRO and the investigator/institution, sponsor and the CRO, and investigator/institution and authority(ies) (where required).

CONCLUSION

Though many stakeholders are involved in clinical research, their roles and responsibilities are clearly mentioned in ICH-GCP. Compliance with the guidelines in every aspect, including documentation, is essential.

Case Exercise 1

An informed consent form (ICF) was submitted to the Ethics Committee for their review and approval along with the prediscussed budget sheet. ICF and budget sheet were approved by the EC, and some of the patients consented on the same ICF.

In the ICF the subject travel reimbursement is written as $50, but in the budget sheet, it has been approved as $100.

Coordinators discover this later when 12 subjects have already consented.

What should be solution now to keep study GCP compliant?

Solution:

The ICF needs to be changed for the statement of patient reimbursement as $100, and along with the budget sheet, it should be submitted to the EC for their acknowledgment.

As both the documents had already been approved by the EC and there has just been an administrative change, there is no need for reapproval and the same should be informed to the EC.

After changing the document, all the subjects should be reconsented and the whole procedure should be documented in source notes.

Case Exercise 2

In a pharma drug trial, the subject had participated in the study. The subject gave his consent, but is illiterate; the subject had been accompanied by his son (blood relative). On behalf of the subject, all the information in the ICF had been written by his son.

Now, once the subject has given their thumb impression (left hand) on the consent form, where should the son sign—in the LAR column or as an impartial witness?

Solution:

The subject's son should sign as an impartial witness as the subject is well, able to give his consent, and can understand the study procedure.

As per the GCP, the LAR comes into play when the subject is unable to give his consent, like infants, comatose patients, etc. So, in this case, the subject's son would be considered an impartial witness.